

From Monocular to Stereo Visual Odometry: *Recovering Metric Scale* on the TUM VI Benchmark

A classical, geometry-based pipeline for fisheye stereo VO — no deep learning, no loop closure.

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BENCHMARK

TUM VI · Room2, Corridor3, Outdoors5
3 sequences · 26 431 frames · 190° fisheye stereo · sub-mm motion capture

STACK

OpenCV 4 · Python 3 · NumPy
Single-thread CPU · seed 42 · no DL · no loop closure

HEADLINE RESULT · GAP-AWARE RPE-TRANSLATION

383_x

average improvement of stereo over monocular VO in local translational drift across 26 431 frames — driven by metric depth recovery from calibrated fisheye stereo.

PROBLEM

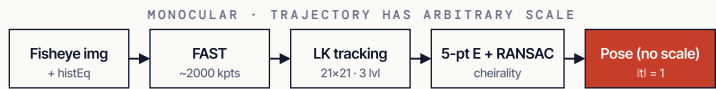
Visual odometry estimates camera ego-motion from images alone. Monocular VO is fundamentally scale-ambiguous: a 1 m trajectory is geometrically indistinguishable from a 1 km one.

On a 0.39 m walk, the monocular estimate integrates to **4565 m** — a scale error of ~10 000×. Calibrated stereo resolves this via metric depth $Z = fB/d$, but TUM VI's **190° fisheye rig** breaks standard rectification, collapsing f_{rect} to ~54 px.

CONTRIBUTIONS

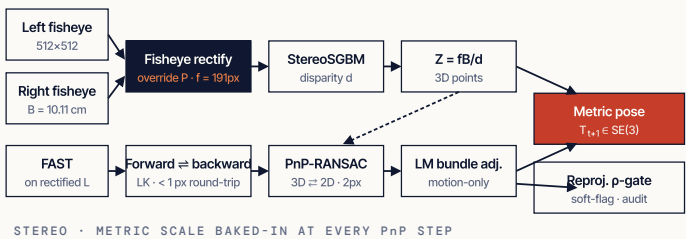
- Monocular pipeline** — FAST + Lucas–Kanade + 5-pt essential matrix on three TUM VI sequences.
- Metric stereo pipeline** — fisheye-aware rectification override, StereoSGBM, PnP-RANSAC + motion-only LM bundle adjustment, forward–backward LK consistency, reprojection-error confidence gate.
- Honest failure handling** — per-frame status logging (OK / TRACK_FAIL / PNP_FAIL / REPROJ_LOW_CONF / STEP_REJECT); hard-skip frames excluded, soft-flags retained & auditable.
- Two evaluation pitfalls corrected** — (a) camera–IMU frame mismatch that inflates raw RPE-rotation by ~18×, (b) multi-minute mocap blackouts on Corridor3 / Outdoors5 that contaminate whole-trajectory ATE.
- Gap-aware protocol** — per-segment ATE, time-thresholded RPE pairs, path-normalised drift.

MONOCULAR PIPELINE



FAST (chosen over Shi–Tomasi: > 2 000 vs. < 200 keypoints on low-brightness fisheye, $\mu \approx 36/255$) feeds LK tracking; pose is recovered up to scale from the 5-pt essential matrix with cheirality. Frames that fail are logged and *excluded*, never frozen.

STEREO PIPELINE



STEREO · METRIC SCALE BAKED-IN AT EVERY PnP STEP

Fisheye rectification fix. Auto-rectification picks $f_{\text{rect}} \approx 54$ px (depth ≤ 5.4 m). Overriding the projection matrices restores $f_{\text{rect}} = 191$ px and reliable depth to ≈ 19 m.

Forward–backward LK. Each feature tracked $t \rightarrow t+1 \rightarrow t$; round-trip residual > 1 px discarded *before* RANSAC — > 90 % fewer PnP failures.

$$\| \text{LK} \downarrow (\text{LK} \uparrow (x_i)) - x_i \| < 1.0 \text{ px} \quad (8)$$

Depth gating. $\delta Z = Z^2 \delta d / (fB)$ reaches 52 % at 20 m. Outdoor preset caps disparity ≥ 1 px and depth ≤ 20 m — *seeing further hurts*.

EVALUATION PITFALLS WE FIX

(A) Camera–IMU frame mismatch. TUM VI mocap is in the IMU body frame, VO in the camera frame — a rigid ~180° offset that **inflates raw RPE-rotation by ~18×**. Correcting collapses error from 47°–53° to **1.0°–1.7°**.

$$\mathbf{T}_{w, \text{cam}}^{\text{gt}} = \mathbf{T}_{w, \text{imu}}^{\text{gt}} \cdot \mathbf{T}_{\text{cam, imu}}^{-1} \quad (13)$$

(B) Multi-minute mocap blackouts. Corridor3 has a ~230 s blackout; Outdoors5 a 12.5 min one. Splitting pairs at $\Delta t > 5$ s and time-thresholding RPE drops Outdoors5 stereo RPE-t from **2.90 m** → **0.037 m**.

TRACKING ROBUSTNESS

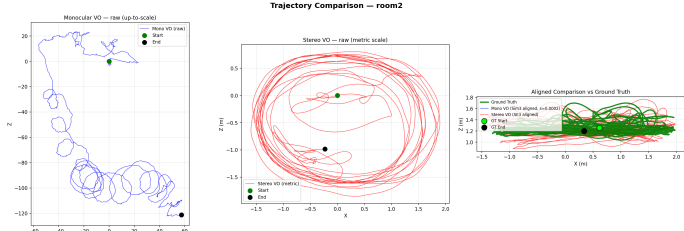
HARD-SKIP RATE · 26 431 FRAMES TOTAL

Sequence	Frames	Mono skip	Stereo skip	Soft flag
Room2	2 882	4.8%	0.0%	0.0%
Corridor3	5 802	4.7%	0.0%	0.3%
Outdoors5	17 747	13.1%	0.1%	0.9%
Total	26 431	10.4%	0.0%	0.7%

TRAJECTORY COMPARISON · EVALUATION OUTPUT § VII-E

Room2 · indoor

2 882 FRAMES · 0.9 M WALK

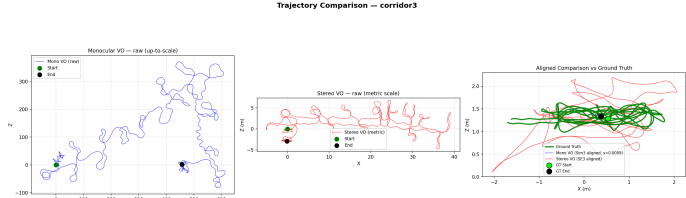


Mono integrates ~1879 m of unit-vector path for a 1.2 m walk. Stereo stays within a metric few-metre range. Aligned view shows stereo and GT covering the same compact region.

ATE **0.29 m** RPE-t **0.020 m** Drift **0.72%**

Corridor3 · indoor

5 802 FRAMES · 230 S MOCAP GAP



Mono produces a spurious ~700 m blob; stereo recovers the corridor as a ~40 m metric trajectory. Aligned view confirms stereo overlays GT through both indoor segments.

ATE **0.024–0.042 m** RPE-t **0.018 m** Drift **1.71%**

HEADLINE METRICS

≤ 0.29_m PER-SEGMENT ATE · ALL SEQUENCES	≤ 0.040_m PER-SEGMENT RPE-T · GAP-AWARE
≤ 1.7° RPE-ROTATION · AFTER FRAME FIX	0.7–4.1% START-END DRIFT · PATH-NORMALISED

QUANTITATIVE COMPARISON

GAP-AWARE METRICS — MONO VS. STEREO

Seq.	Mode	ATE _{seg} (m)	RPE-t (m)	RPE-r (°)	Drift %
Room2	mono	1.23	9.83	46.57	0.49
	stereo	0.29	0.020	1.10	0.72
Corridor3	mono	0.72–0.77	8.03	53.61	1.18
	stereo	0.024–0.042	0.018	1.09	1.71
Outdoors5	mono	0.73–0.91	7.78	62.96	0.26
	stereo	0.15–0.17	0.037	1.62	4.10

Mono drift after Sim(3) alignment (raw unit-vector path: 1879 / 707 / 4565 m). Stereo Drift% on par with ORB-SLAM2 KITTI (~1%) and within 3–5% typical of classical VO without loop closure.

ABLATION ON OUTDOORS5

CUMULATIVE · GAP-AWARE

Configuration	ATE	RPE-t	RPE-r	Drift %
Initial outdoor preset	0.59	0.075	5.49	13.31
+ GT in camera frame	0.42	0.061	2.27	6.06
+ tight depth + LM bundle adj.	0.28	0.046	1.94	4.81
+ Forward–backward LK consistency	0.16	0.037	1.62	4.10

DISCUSSION & FUTURE WORK

- Mono ATE is misleading** — Sim(3) Umeyama absorbs all scale error; RPE-t (7.8–9.8 m mono vs. 0.02–0.04 m stereo) and the 5-pt degeneracy under near-pure rotation (RPE-r 47°–63°) are the honest cross-modal failures.
- Next:** local bundle adjustment, DBow2 loop closure, 200 Hz IMU pre-integration — building on the calibrated geometry established here.

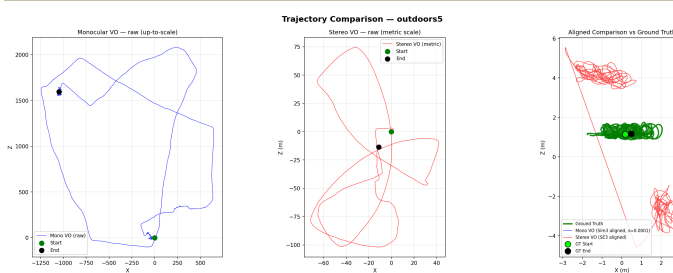
KEY REFERENCES

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- Hartley & Zisserman. *Multiple View Geometry in CV*, 2nd ed., Cambridge, 2004.
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■ Mono VO ■ Stereo VO ■ Ground truth

Outdoors5 · 1.1 km loop

17 747 FRAMES · 749 S MOCAP GAP



Mono produces a ~4.5 km spurious shape; stereo recovers the outdoor loop topology at metric scale. Aligned view shows the indoor GT clusters (start & end) — the 749 s outdoor traversal has no GT and is the source of all visible offset.

ATE **0.15–0.17 m** RPE-t **0.037 m** Drift **4.10%**